Fewer molecules. Deeper experimental data.

AI may reduce candidate counts while increasing characterization per molecule

From molecule throughput to data density: a potential driver of life science tools demand

Existing data: staged screening

1,000 × 1 initial label

Illustrative early single-task dataset

Antibody	Binds?
Ab-001	Yes
Ab-002	No
Ab-003	Yes
Ab-004	Yes
Ab-005	No
Ab-006	Yes

Binding first; deeper tests for selected hits

Multi-assay characterization already exists

AI: richer, linked training data

300–1,000 antibodies · Up to 9 label types

Measure selected properties on shared molecules; fill the information gaps

Antibody	KD	kon	koff	expr	Tm	agg	poly	epi	cell
Ab-001	120	7200	1800	8400	5100	75	3200	9000	430
Ab-002	8100	640	5400	2200	9500	1600	4100	500	6900
Ab-003	2600	9200	410	6500	3100	7900	950	4800	8300
Ab-004	9800	3100	7400	1200	6800	380	5500	1900	9100
Ab-005	540	4700	8800	3600	2000	6200	10000	710	5700
Ab-006	6300	1400	3000	9900	4600	8700	180	7600	2400

Learn binding, expression, stability and functional relationships

Synthetic values: 1–10,000. Darker = larger; no biological units or quality ranking.

Reuse existing data

Normalize formats, conditions and access
Better training may need no new experiment

Stage 0: fill data gaps

Add missing assays and biological diversity
Create reusable model-training datasets

Active learning: measure selectively

Prioritize informative molecules and tests
Limit coverage of costly measurements

The nine labels: what each property measures

KD	Equilibrium binding strength	**kon**	How quickly binding forms	**koff**	How quickly the complex separates
expression	Antibody yield from a cell system	**Tm**	Thermal unfolding transition	**aggregation**	Molecules forming larger clusters
polyreactivity	Binding to multiple non-targets	**epitope**	The binding site on the target	**cell potency**	Functional activity in living cells

Illustrative counts. Nine label types are not nine independent assays; richer data need not come at unchanged cost.

Sources: Jain et al., 2017; Arsiwala et al., 2025; van Niekerk et al., 2026. Mechanism illustration; definitions and links in SOURCES.md.

Experimental data density and the LST ecosystem

Stage 0: biological data production creates a wet-lab use case

Learning developability requires comparable assay data and an integrated tool workflow

Stage 0: biological data production

Continuous input to training and iteration

Binding kinetics
KD / kon / koff

Expression & physicochemical properties
expression / Tm / aggregation

Specificity & cell function
polyreactivity / epitope / cell potency

Training records: sequence, target, conditions, raw curves, replicates, batches and failures

Train / improve → **Design / validate**

Assays inform models

Labels → assays → purchases

Tool revenue needs a bridge from label coverage to workflow volume and unit consumption

Construction & expression

DNA synthesis, vectors, expression, purification

Expression labels + samples for other assays

Count shared sample preparation once

Protein & functional testing

SPR / BLI, binding curves, epitope binning

Target proteins, sensors / chips, buffers

KD / kon / koff can share one kinetic run

Cells & culture systems

Functional cells, media, transfection, expansion

Cell potency; selected polyreactivity assays

Record cell model, target level and conditions

Measurement & analysis

SEC / HIC, nanoDSF, LC-MS, flow cytometry

Stability, aggregation, characterization / QC

Columns are reused; model instrument CAPEX separately

Sample handling, labware & automation

Plates, tips, tubes, filtration / beads, liquid handling

Across all workflows; miniaturization, reuse and scheduling alter net consumption

ACROBiosystems: target proteins, functional cells & characterization

AIDD-Ready offering exists; paid orders and repeat purchases still need verification

NanoMicro: analytical separation & lab purification materials

Track sample runs, column life and ASP; manufacturing resin demand has a longer lag

LST purchases ≈ Σ [workflow executions × net purchased inputs per run × ASP]

Instrument CAPEX is separate. Product alignment does not establish AI orders or revenue. Sources: Ginkgo, Sartorius and company websites.